

A Radius-One Obstruction in Reconfiguring Nowhere-Zero Flows toward Five-Cycle Double Covers

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Status and AI-use disclosure. The standard five-cycle double cover conjecture remains open. This manuscript proves a short packing-to-switch lemma and gives exact finite counterexamples only to a proposed radius-one reconfiguration lemma. It proves neither FiveCDC nor its negation.

OpenAI Codex agents using GPT-5-series models, under human direction, formulated the auxiliary question and the packing-to-switch lemma, found the finite examples, wrote the search and both checking programs, performed the computations, drafted the proof and this manuscript, and conducted an AI-assisted literature search. The “independent checker” is independent of the discovery implementation, not independent of AI. No human author, referee, or cited researcher is represented as having verified these results. Before submission, a human author must inspect every proof, independently rerun and preferably reimplement the enumerations, audit the graph and cover certificates, review the literature, and accept ordinary scholarly responsibility.

Abstract

Hušek and Šámal recently gave a component-parity condition on a nowhere-zero $\mathbb{F}_2^3$ -flow which is equivalent, after choosing the flow, to the existence of a five-cycle double cover. Cranston, Li, Su, Wang, and Xu study reconfiguration of nowhere-zero flows by changes supported on cycles. We test the direct radius-one synthesis of these ideas.

First we prove a human-checkable bridge. If one value-class matching of a cubic $\mathbb{F}_2^3$ -flow admits two edge-disjoint joins in its complement, then an explicitly determined even support, and therefore a finite sequence of legal simple-cycle switches, reaches a flow satisfying the Hušek–Šámal condition. Connectedness of that support is exactly what makes one switch sufficient.

We then give an exact 26-vertex obstruction to radius one. The graph is simple, cubic, cyclically 4-edge-connected, has girth five, and is not Tait-colourable. One displayed value class already admits two edge-disjoint boundary joins. Nevertheless, for the displayed nowhere-zero flow, a standard-library checker exhausts all 9,213 simple cycles and all 1,485 legal cycle–value switches; none reaches the component condition. The packing construction splits into two circuit components, and switching both reaches the condition. Thus the displayed flow has exact distance two. An independently checked five-cycle double cover of the same graph confirms that this is not a counterexample to FiveCDC.

For calibration we record a cyclically 4-edge-connected, girth-five, Tait-colourable 40-vertex radius-one obstruction, and a 60-vertex positive control repaired by one 25-edge cycle. The latter is also a literal instance of the packing-to-switch lemma. All finite claims are accompanied by solver-independent, certificate-based checking code. Possible novelty is stated conservatively and remains subject to specialist prior-art review.

1 Scope and relation to prior work

A *five-cycle double cover* (FiveCDC) of a finite graph is an indexed list $C_0, \dots, C_4$ of Eulerian edge-subsets such that every edge lies in exactly two members. Eulerian means even degree at every vertex; neither connectedness nor 2-regularity is required. Empty members are allowed, so this is the usual “at most five” convention. Only the standard, unoriented conjecture is considered here.

Hušek and Šámal state FiveCDC as Conjecture 1.9 and give an equivalent flow-selection formulation as Conjecture 3.19 in [2]. In particular, their Theorem 3.16 supplies the component-parity criterion used below. Thus FiveCDC remains open, and the remaining universal issue is to select a suitable nowhere-zero $\mathbb{F}_2^3$ -flow.

Cranston, Li, Su, Wang, and Xu define adjacency of nowhere-zero flows by requiring their difference to be supported on a cycle [1]. In the exponent-two group $\mathbb{F}_2^3$, a difference supported on a simple cycle has one constant nonzero value around that cycle. The switches below are precisely this special case, with legality made explicit so that no zero edge is created.

Our contribution has deliberately narrow scope:

1. a direct proof that a two-join packing yields a finite legal switch sequence to a Hušek–Šámal-good flow;
2. an exact strict 26-vertex countermodel to the assertion that one simple-cycle switch always suffices even when a value class already packs, together with an exact distance-two path and a positive FiveCDC certificate; and
3. one 40-vertex negative and one 60-vertex positive control.

No reduction from FiveCDC to a universal radius-one assertion is claimed here, and no statement about orientable cycle double covers is made.

2 Flows, component defects, and switches

Put $W = \mathbb{F}_2^3$, written additively, and identify W with $\{0, \dots, 7\}$ by binary coordinates. Let G be a finite simple cubic graph. A map

$$f : E(G) \longrightarrow W - \{0\}$$

is a nowhere-zero flow if

$$\sum_{e \ni v} f(e) = 0 \quad (v \in V(G)).$$

At a cubic vertex the three incident values are nonzero and pairwise distinct. Consequently each value class

$$M_a(f) = \{e : f(e) = a\}, \quad a \neq 0,$$

is a matching.

For a nonzero linear functional $\mu \in W^*$, identified with its binary dot-product vector, define

$$\begin{aligned} F_\mu(f) &= \{e : \mu(f(e)) = 1\}, \\ K_\mu(f) &= \{e : \mu(f(e)) = 0\} = E(G) - F_\mu(f). \end{aligned} \tag{2.1}$$

For $r \in W$ with $\mu(r) = 1$, let

$$Z_r(f) = \partial M_r(f)$$

be the endpoints of the matching $M_r(f)$. Define

$$d_\mu(f) = \# \{Q \in \text{Comp}(V(G), K_\mu(f)) : |Q \cap Z_r(f)| \text{ is odd}\}. \tag{2.2}$$

Lemma 2.1 (affine-value independence). *The component parities in (2.2) do not depend on the choice of r satisfying $\mu(r) = 1$.*

Proof. Fix a component Q of $K_\mu(f)$. The parity of $|Q \cap Z_r(f)|$ equals the parity of the number of r -valued edges in $\delta(Q)$, since an internal matching edge contributes two endpoints. Every edge in $\delta(Q)$ has μ -value one.

Write $\ker \mu = \{0, p, q, p + q\}$, so the four affine values are $r, r + p, r + q, r + p + q$. Let their boundary multiplicities modulo two be n_0, n_p, n_q, n_{p+q} . Summing the flow equations on Q gives

$$n_0 r + n_p(r + p) + n_q(r + q) + n_{p+q}(r + p + q) = 0.$$

Comparing coefficients in the basis (r, p, q) gives

$$n_0 + n_p + n_q + n_{p+q} = 0, \quad n_p + n_{p+q} = 0, \quad n_q + n_{p+q} = 0.$$

Hence all four multiplicities are equal. This proves the claim. $\square$

Definition 2.2. The flow f is *H-S-good* if $d_\mu(f) = 0$ for some $\mu \neq 0$. Its defect profile is

$$d(f) = (d_1(f), \dots, d_7(f)).$$

By Theorem 3.16 and Observation 3.15 of [2], an H-S-good flow produces a standard FiveCDC. Their Conjecture 3.19 asserts that every bridgeless cubic graph has some H-S-good flow and is equivalent to FiveCDC.

Definition 2.3 (legal simple-cycle switch). Let $a \in W - \{0\}$, and let C be a simple cycle of G such that $C \cap M_a(f) = \emptyset$. Define

$$f^{a,C}(e) = \begin{cases} f(e) + a, & e \in E(C), \\ f(e), & e \notin E(C). \end{cases} \quad (2.3)$$

The cycle equation preserves flow conservation, and $C \cap M_a(f) = \emptyset$ is exactly the condition preventing a zero value. We write $\text{dist}_{\text{HS}}(f)$ for the minimum number of legal simple-cycle switches needed to reach an H-S-good flow, when such a sequence exists.

3 A human packing-to-switch lemma

For an edge set J , write ∂J for the set of vertices having odd degree in J . A *T-join* is an edge set J with $\partial J = T$.

Theorem 3.1 (packing-to-switch). *Fix $a \neq 0$ and $\mu(a) = 1$. Put*

$$F = F_\mu(f), \quad M = M_a(f), \quad J_0 = F - M.$$

Suppose $G - M$ contains two edge-disjoint ∂M -joins J_1, J_2 . Then switching a on

$$C = J_0 \triangle J_1 \quad (3.1)$$

produces an H-S-good flow. The support C is a disjoint union of legal simple cycles, so this operation is a legal finite sequence of simple-cycle switches. If C is connected, one switch suffices.

Proof. The binary characteristic function $\mu \circ f$ is a flow, so F is Eulerian. Because $M \subseteq F$,

$$\partial J_0 = \partial(F - M) = \partial M.$$

Thus J_0 and J_1 have the same boundary, and $C = J_0 \triangle J_1$ is Eulerian. Both avoid M , hence so does C .

In a cubic graph every vertex has degree zero or two in an Eulerian edge set. Every nonempty connected component of C is therefore a simple cycle. Switch a on all these components. Since $\mu(a) = 1$, the new affine support is

$$\begin{aligned} F \triangle C &= F \triangle ((F - M) \triangle J_1) \\ &= M \cup J_1. \end{aligned} \tag{3.2}$$

The union is disjoint because $J_1 \subseteq E(G - M)$.

The matching M_a itself is unchanged: the support avoids M_a , and an edge with old value different from a cannot acquire value a after adding a , since that would force its old value to be zero. Consequently every component switch remains legal when the components are performed sequentially.

Finally, J_2 avoids both M and J_1 , so by (3.2) it lies in the new zero-side K_μ . Restricting the equality $\partial J_2 = \partial M$ to any component Q of this zero-side and applying the handshaking lemma gives

$$|Q \cap \partial M| \equiv 0 \pmod{2}.$$

By Lemma 2.1, this is the H-S condition for μ . □

Remark 3.2. The theorem exposes the radius-one gap. A packing gives an Eulerian switch support canonically, but that support may have several cycle components. The theorem guarantees a finite reconfiguration sequence; it guarantees one step only when the support is connected.

4 The direct radius-one assertion

The most direct strengthening suggested by Theorem 3.1 is:

For every simple cubic cyclically 4-edge-connected graph and every nowhere-zero W -flow f , either f is H-S-good or one legal simple-cycle switch from f is H-S-good.

This statement concerns the current H-S component condition, not a fixed five-cover and not an older all-value-class nonpacking condition. The following exact countermodel refutes it.

5 An exact strict 26-vertex countermodel

Graph6 orders the edges lexicographically by second endpoint and then by first endpoint. Consider the graph

Y??G@c00GC??P??Ap??i_A??CIC???IC_C?@??H0??A??B_G????D?_

and the following flow values in that edge order:

$$\begin{aligned} (5, 1, 5, 4, 4, 4, 7, 1, 6, 7, 3, 1, 6, 4, 3, 2, 3, 2, 6, 1, \\ 2, 5, 5, 2, 7, 6, 3, 2, 5, 7, 1, 1, 7, 6, 5, 3, 1, 7, 6). \end{aligned} \tag{5.1}$$

Theorem 5.1 (strict radius-one countermodel). *The graph and flow in (5.1) have the following properties.*

1. *The graph is simple, cubic, cyclically 4-edge-connected, has girth five, and has no Tait 3-edge-colouring.*
2. *The displayed values form a nowhere-zero W -flow with profile*

$$(6, 6, 4, 6, 2, 4, 4).$$
3. *Its value-4 class packs two edge-disjoint boundary joins in its complement.*
4. *The graph has 9,213 simple cycles and exactly 1,485 legal cycle-value switches from this flow. None is H - S -good. The minimum defect at radius at most one is two.*
5. *The graph has the explicit standard FiveCDC given below.*
6. *The displayed flow has exact H - S distance two.*

FiveCDC certificate. Encode a pair of cover coordinates by a five-bit integer. In graph6 edge order, the labels are

$$\begin{aligned} (3, 20, 5, 17, 3, 5, 6, 3, 3, 5, 10, 6, 3, 5, 3, 6, 5, 9, 10, 12, \\ 6, 18, 24, 12, 20, 6, 18, 6, 20, 18, 6, 6, 18, 20, 6, 5, 5, 6, 3). \end{aligned} \quad (5.2)$$

Every label in (5.2) has Hamming weight two. For each of the five bit positions and each vertex, an even number of incident labels contain that bit. Thus the five bit-coordinate edge sets are Eulerian and every edge belongs to exactly two of them.

Exact distance-two certificate. Starting from (5.1), perform

$$\begin{aligned} a = 4 \quad C &= (0, 1, 2, 8, 9, 20, 22, 24, 25, 26, 30, 31, 33), \\ a = 4 \quad C &= (6, 12, 14, 15, 34, 35). \end{aligned} \quad (5.3)$$

Both supports are simple cycles and avoid the current value class being added. The successive profiles are

$$(4, 6, 4, 2, 4, 4, 4), \quad (4, 6, 4, 0, 6, 4, 4). \quad (5.4)$$

Thus (5.3) gives an H - S -good flow after two steps.

Packing certificate. The value-4 matching is

$$M_4 = (3, 4, 5, 13).$$

The two edge sets

$$\begin{aligned} J_0 &= (1, 14, 15, 18, 20, 21, 26, 28, 29, 30, 31, 32, 35, 37, 38), \\ J_1 &= (0, 2, 6, 7, 8, 9, 12, 16, 17) \end{aligned}$$

are disjoint, avoid M_4 , and both have boundary ∂M_4 . The even support furnished by Theorem 3.1 has exactly the two components in (5.3). Hence this certificate explains the distance-two repair while proving that packability alone does not force a radius-one neighbour.

Certificate proof of Theorem 5.1. The supplied standard-library checker decodes graph6 from first principles. It verifies distinct edges, degree three, connectedness, the 26 vertex flow equations, and all 39 nonzero values. It examines every deletion of one, two, or three edges and rejects a deletion as a cyclic cut only after constructing all components and confirming that at least two contain cycles. It independently computes girth and exhaustively backtracks over proper three-edge-colourings.

For completeness of the radius-one enumeration, the checker performs a depth-first enumeration of simple cycles. The minimum vertex of a cycle is its unique allowed starting vertex. The two possible traversal directions are quotiented by retaining the direction whose second vertex is smaller than its final vertex. Hence each undirected simple cycle is generated exactly once. For each generated cycle and each $a \in W - \{0\}$, the checker tests legality directly and, when legal, recomputes the full seven-entry defect profile from graph components. The resulting exact totals are 9,213 cycles, 1,485 legal switches, and zero clean neighbours. This proves $\text{dist}_{\text{HS}}(f) \geq 2$. Direct verification of (5.3)–(5.4) proves $\text{dist}_{\text{HS}}(f) \leq 2$.

Finally, the checker verifies the matching and both join boundaries, and tests every weight and every vertex-coordinate parity in (5.2). These finite checks prove all six assertions. $\square$

Remark 5.2. Theorem 5.1 is a counterexample only to the radius-one auxiliary statement. Its explicit FiveCDC proves that it is not a FiveCDC counterexample.

6 Forty- and sixty-vertex controls

Two controls separate radius-one failure from the existence of a cover and from the packing-to-switch mechanism.

6.1 A Tait-colourable 40-vertex negative control

There is a simple cubic, cyclically 4-edge-connected graph of girth five with a nowhere-zero flow having profile

$$(6, 8, 10, 4, 10, 6, 6). \quad (6.1)$$

The checker exhausts 941,438 simple cycles and 48,544 legal cycle-value switches. None is H-S-good, and the minimum defect at radius at most one is two. In contrast to Theorem 5.1, this graph has a displayed Tait three-edge-colouring; its three colour-pair 2-factors give a three-cycle double cover. This is therefore a clean negative control for the reconfiguration assertion, not for FiveCDC.

6.2 A 60-vertex positive control and the packing lemma

The 60-vertex control is simple cubic and cyclically 4-edge-connected. Its starting profile is

$$(8, 8, 4, 6, 6, 8, 6). \quad (6.2)$$

Switching $a = 7$ on the 25-edge simple cycle

$$C = (1, 2, 8, 11, 18, 22, 24, 26, 27, 30, 31, 33, 35, 37, 39, 40, \\ 42, 43, 45, 62, 65, 67, 73, 74, 75) \quad (6.3)$$

gives profile

$$(10, 6, 6, 8, 8, 6, 0). \quad (6.4)$$

Thus this state is repaired in one step.

Table 1: The three principal exact computations. “Legal” counts cycle–value pairs, not merely cycles.

order	cycles	legal	initial minimum	radius-one minimum	outcome
26	9,213	1,485	2	2	packs; no one-switch repair
40	941,438	48,544	4	2	no repair; Tait control
60	—	—	4	0	displayed one-switch repair

The repair also instantiates Theorem 3.1. For the value-7 matching, two edge-disjoint ∂M_7 -joins in $G - M_7$ are

$$\begin{aligned}
J_1 &= (1, 2, 13, 16, 18, 20, 26, 32, 35, 44, 46, 49, 50, 53, 57, 60, 61, \\
&\quad 62, 63, 64, 69, 71, 72, 73, 75, 81, 83), \\
J_2 &= (3, 6, 7, 10, 12, 14, 17, 19, 22, 24, 27, 29, 34, 37, 40, 41, 43, \\
&\quad 45, 47, 48, 55, 56, 74, 80, 82, 85, 86, 87, 88).
\end{aligned} \tag{6.5}$$

With $F = F_7(f)$, the checker verifies

$$C = (F - M_7) \triangle J_1, \quad J_2 \cap (M_7 \cup J_1) = \emptyset. \tag{6.6}$$

Equations (6.3)–(6.6) are therefore a literal connected-support case of the human lemma.

7 Verification architecture

The principal checker is

`scratch/verify_husek_samal_packable_one_switch_countermodel.py`.

It uses only the Python standard library. Its graph6 decoder, flow checker, component-defect routine, simple-cycle generator, switch routine, cyclic-cut enumeration, girth computation, Tait-colouring backtracker, and FiveCDC checker are implemented locally. No SAT solver or opaque mathematical library is used.

The 40-vertex negative control is checked by the separately written

`scratch/verify_husek_samal_one_switch_boundary.py`.

The 60-vertex construction has a second checker,

`scratch/verify_fano_order60_flow_repair.py`.

It reconstructs the graph from two 32-vertex pole recipes, verifies the one-switch repair and the two joins, and separately follows a 13-switch path to the flow induced by an explicit FiveCDC. The two programs share no imported project module.

The frozen artifact hashes at the time of this draft are:

SHA-256	file
9bdc3a65f6dc07224f8a20c79e45e9085c9f73c55ead745ac64127dec76607	packable-state checker
ac51aaf266178eb764fcada85b25e17dc2c18c129413abb51a2b309513ea415d	boundary checker
09f9bdc797948b49cf6b7a3080160e07111dad718f0e07ae00a2a9c0958b9a6e	independent 60-vertex replay
5cd280e730e0dc336681c3768945b7c691e023cb4f957a7408ea306597053f45	packable order-26 retained state
6e1dc5fb3b290fe8364559844ed419ee6dda1ec974b2ad3aebfd21442c8fe6aa	order-40 retained state

From the project root, the exact replay commands are

```
python3 scratch/verify_husek_samal_packable_one_switch_countermodel.py
python3 scratch/verify_husek_samal_one_switch_boundary.py
python3 scratch/verify_fano_order60_flow_repair.py
```

8 What remains open

The finite countermodels show only that radius one is not a universal domination bound for H-S-good flows, even in a strict reduced-looking class. They leave open all of the following:

1. whether every nowhere-zero $\mathbb{F}_2^3$ -flow can reach an H-S-good flow by some finite legal cycle-switch sequence;
2. whether a larger universal radius exists;
3. whether every bridgeless graph has some initially H-S-good flow, the equivalent formulation of FiveCDC in [2]; and
4. the standard FiveCDC conjecture itself.

The 26-vertex state has exact distance two, but this is one instance, not evidence of a universal radius-two theorem. It also shows sharply that the finite sequence in the packing-to-switch lemma cannot always be replaced by one circuit. The lemma supplies no universal packing theorem.

9 Novelty and responsibility boundary

The cited papers establish the H-S component criterion and the general cycle-supported reconfiguration framework. We make no novelty claim for either. The elementary packing-to-switch bridge, the exact radius-one formulation, and the three finite certificates may be new, but that assessment is provisional. The search and literature review were AI-assisted and not exhaustive in the scholarly sense. Priority and publishability require review by specialists in cycle double covers, nowhere-zero flows, and flow reconfiguration.

The human part of Theorem 3.1 is fully displayed. The finite theorems reduce to explicit data and short, inspectable exhaustive programs, but those programs and this prose were generated with AI assistance. Reproducibility is not peer review, and implementation independence is not human independence.

References

- [1] Daniel W. Cranston, Jiaao Li, Bo Su, Zhouningxin Wang, and Ningyan Xu. Reconfiguration of nowhere-zero flows, 2026.
- [2] Radek Hušek and Robert Šámal. Exponentially many circuit double covers, 2026.